Lab 4

System Modeling Using UML – Use Case and Sequence Diagrams

Part A – System Understanding & Use Case Identification

Task A1: Identify Actors

Each team identifies:

- Primary actors
- Secondary actors (if any)

Deliverable:

- Actor list with brief description

Primary Actors

1. **Customer (End User)**
   The customer is the main user of the system who registers, logs in, browses products, adds items to the cart, and places orders.
2. **Administrator (Admin)**
   The admin manages the system by adding, updating, or deleting products and monitoring customer orders.

Secondary Actors

1. **Payment Gateway System**
   An external system that processes online payments securely during checkout.
2. **Database System**
   Stores user data, product details, and order information. It supports system operations but does not directly interact with users.

Task A2: Identify Use Cases

From SRS / user stories:

• Identify 6–8 major use cases

Examples (generic):

• User Registration

• Login

• Search Information

• Update Profile

• Generate Report

• Logout

Deliverable:

• Use Case list with short description

Based on the SRS and user stories of the E-Commerce Website, the following major use cases are identified:

1. User Registration
   Allows a new user to create an account by providing required details such as email and password.
2. User Login
   Allows a registered user to log in securely using valid credentials.
3. Browse Products
   Enables users to view available products categorized by type.
4. View Product Details
   Allows users to see detailed information about a selected product, including price and description.
5. Add to Cart
   Enables users to add selected products to their shopping cart for purchase.
6. Place Order
   Allows users to confirm their cart items and generate an order.
7. Make Payment
   Enables users to complete the purchase through an online payment gateway.
8. Manage Products (Admin)
   Allows the administrator to add, update, or delete product information.

Part B – Use Case Diagram

Task B1: Draw Use Case Diagram

Teams create a UML Use Case Diagram showing:

- Actors
- Use cases
- Relationships:
- Association
- «include» / «extend» (where applicable)

Rules:

- System boundary must be clearly shown
- Avoid implementation details

Deliverable:

- One complete Use Case Diagram (hand-drawn or tool-based)

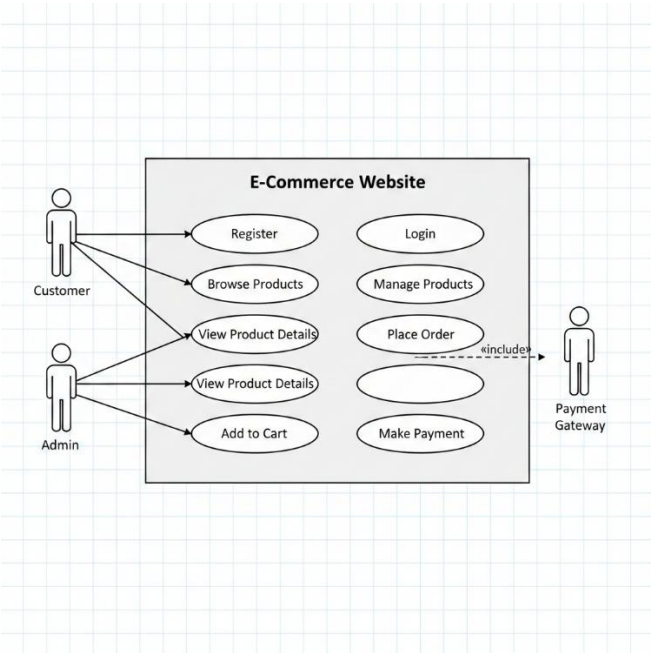

Figure 1: UML Use Case Diagram for Simple E-Commerce Website

Part C – Use Case Selection for Sequence Modelling

Task C1: Select Use Cases

Each team selects 2–3 important use cases for sequence modeling.

Examples:

- User Registration
- Login
- Place Order / Book Appointment / Submit Request

Deliverable:

- Selected use case names with justification

Selected Use Cases

1. **User Registration**
   This use case is selected because it represents the first interaction between a new user and the system. It involves validation and data storage, making it suitable for modelling message flow between the user interface and system components.
2. **User Login**
   This use case is important as it involves authentication and verification processes. It demonstrates interaction between the user, system, and database for credential validation.
3. **Place Order**
   This use case is selected because it is a core business function of the e-commerce system. It involves multiple steps such as cart verification, order creation, and payment processing, making it ideal for detailed sequence modelling.

Part D – Sequence Diagrams

Task D1: Identify Objects

For each selected use case, identify:

• Actor

• Boundary object (UI)

• Control object (logic)

• Entity objects (data)

1. Use Case: User Registration

- Actor: Customer
- Boundary Object (UI): Registration Page
- Control Object (Logic): Registration Controller
- Entity Objects (Data): User, User Account

2. Use Case: User Login

- Actor: Customer
- Boundary Object (UI): Login Page
- Control Object (Logic): Authentication Controller
- Entity Objects (Data): User, User Credentials

3. Use Case: Place Order

- Actor: Customer
- Boundary Object (UI): Checkout Page
- Control Object (Logic): Order Controller
- Entity Objects (Data): Cart, Order, Payment, Product

Task D2: Draw Sequence Diagrams

For each selected use case, create a Sequence Diagram showing:

• Lifelines

• Messages (method calls)

• Activation bars

• Response messages

Rules:

• Follow logical flow

• Use meaningful message names

• One diagram per use case

Deliverables:

• Minimum 2 sequence diagrams

• Clearly labelled participants and messages

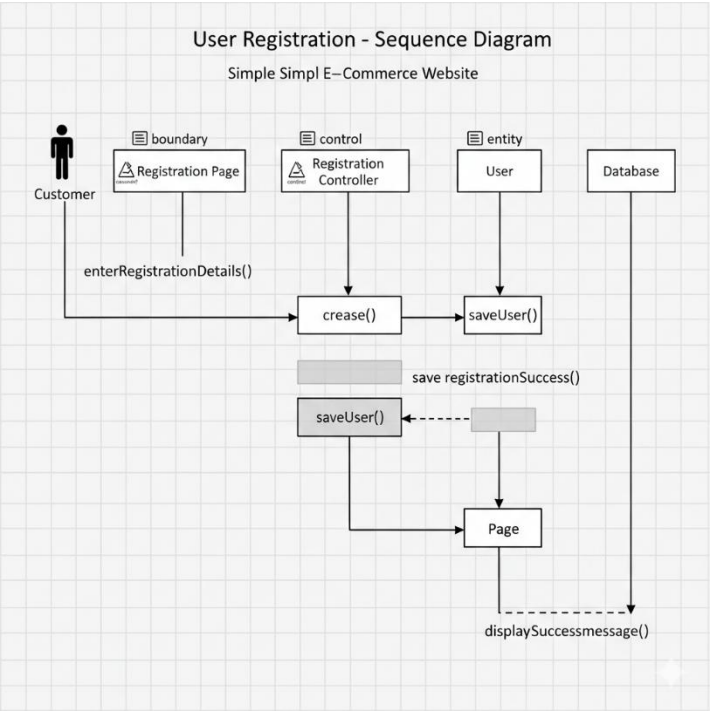

Figure 2: UML Sequence Diagram for User Registration

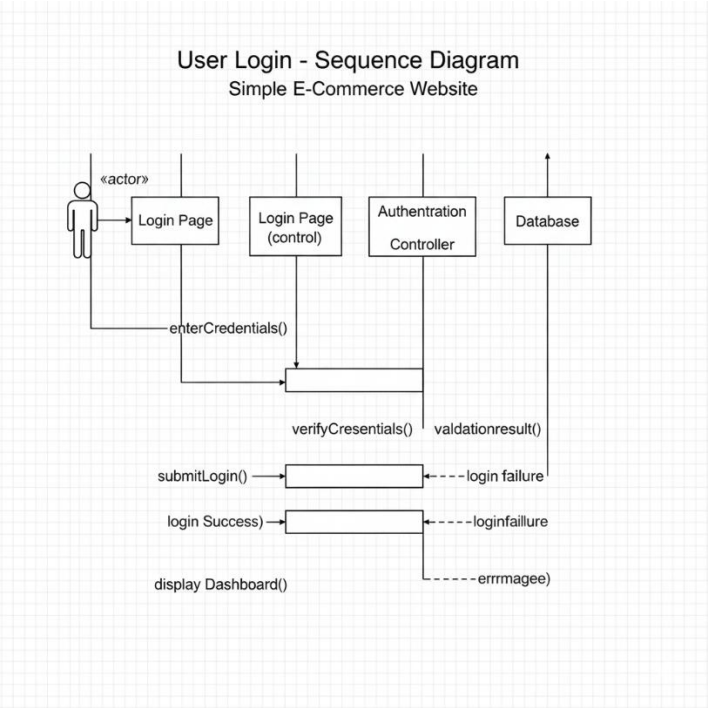

Figure 3: UML Sequence Diagram for User Login

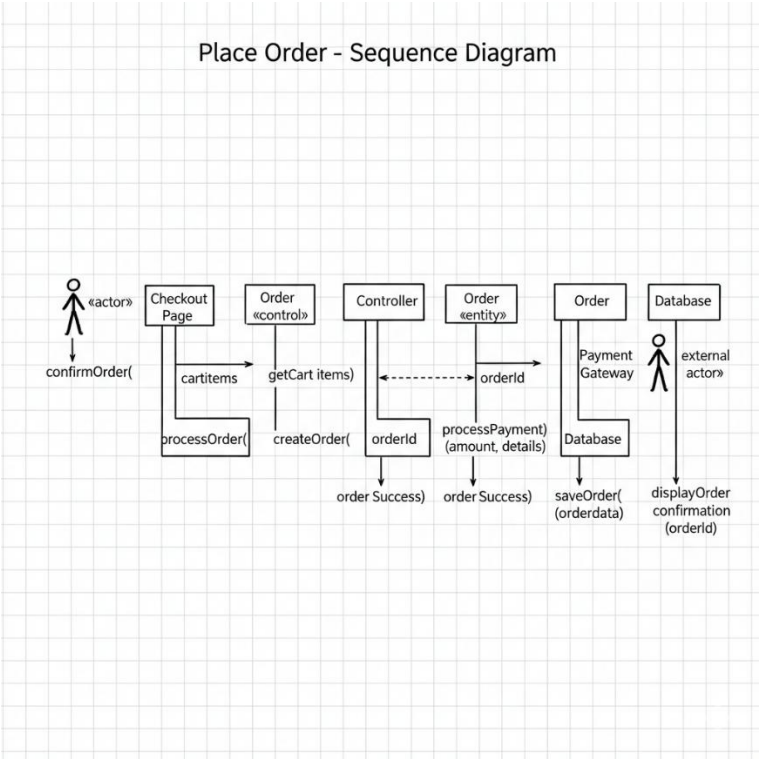

Figure 4: UML Sequence Diagram for Place Order

Conclusion

In this lab, we analysed the E-Commerce Website system and identified its primary and secondary actors along with major use cases. A complete UML Use Case Diagram was created to represent the interactions between users and the system within a clearly defined system boundary. Important use cases were selected and further modelled using Sequence Diagrams to illustrate detailed message flow between actors, boundary objects, control objects, and entity objects.

This lab helped in understanding how UML modelling techniques represent system behaviour logically without including implementation details. The use of use case and sequence diagrams improves clarity, enhances requirement understanding, and supports effective system design and documentation.

Team Members

Siddeshwarprasad K R (1RUA24CSE0454)

Shreyas C P (1RUA24CSE0443)

Nikhil Savanth K M (1RUA24CSE0282)

Ullas Gowda R (1RUA25CSE7017)